

# ChE 311 Transport phenomena I

## General information

3 credit hours

Instructor: Ying Liu

Research interest: complex fluids, microfluidics, nanotechnologies, soft matter physics

Office: EIB 152

Telephone:

Email: [liuying@uic.edu](mailto:liuying@uic.edu)

Teaching assistant: Yuechen Gao

Textbook: R.B.Bird, W.E. Stewart & E. N. Lightfoot, Transport Phenomena, 2ed Ed., Wiley (2002)

Lectures: MWF, 11:00 - 11:50 am

Location: Lecture Center C Room C001

Ying Liu Office Hours: W 1:00 pm – 2:00 pm

Th 1:00 pm – 2:00 pm

Yuechen Gao Office Hours: TTh 2-3pm

Prerequisite(s): credit or concurrent registration in CHE 210, MATH220, CHE205.

## Course description

Momentum transport phenomena in chemical engineering. Fluid statics. Fluid mechanics; laminar and turbulent flow; boundary layers; flow over immersed bodies.

Examples

Industry: mixing, coating...

Transportation: airplane, vehicle design...

Sports

Life: bullet-proof jacket, quicksand...

Animal: penguins, fish...

Weather: storms...

Environment: iceberg, dam...

## Methods of evaluation

Distribution of final grade: Homework: 10%

Midterm exam: (each 20%, total 2×20%)

Quiz: (each 5%, total 4×5%)  
Final exam (optional): 30%

Score distribution: A: ~85 and above  
B: ~60-84  
C: ~45-59  
D: ~30-44  
F: ~less than 30

#### Exam schedule

| Exam | Quiz 1 | Quiz 2  | Quiz 3 | Midterm 1 | Quiz 4  | Quiz 5  | Midterm 2 |
|------|--------|---------|--------|-----------|---------|---------|-----------|
| Week | Week 3 | Week 5  | Week 7 | Week 9    | Week 11 | Week 13 | Week 15   |
| Date | Sep. 9 | Sep. 23 | Oct. 7 | Oct. 21   | Nov. 4  | Nov. 18 | Dec. 2    |

Homework is due weekly on Friday (before 5:00 pm). TA should return the evaluated homework before the following Friday (before 5:00 pm).

#### Topical outline

Week 1-3: Transport coefficients (Chap. 1, 9, 17)

Fourier's law of heat conduction

Fick's law of binary diffusivity

Newton's law of viscosity

Introduction and mathematical preliminaries (Appendix A)

Scalars, vectors, and tensors—operations involving them

Week 3-5: Shell Momentum balances and velocity distribution in laminar flow (Chap. 2)

Week 5-7: Conservation equations (Chap. 3)

Differential equations for mass, linear momentum, and energy

Week 7-10: Macroscopic balances (Chap. 7)

Integral balance equations for mass, linear momentum, and energy

Week 10-12: Introduction of turbulent flow (Chap. 5)

Week 12-13: Friction factors (Chap. 6)

Week 13-15: Velocity distributions with more than one independent variable (Chap. 4)

Time dependence

Stream function

### **Course policies**

Homework: Solution manual can be used to check the correctness after homework has been done independently. Group discussion is allowed. Points will be deducted if homework is turned in late. Exception can be made with appropriate reasons with proof.

Exam: All the exams (including quiz, mid-term and final) are closed-book exams. Tables will be provided by the instructor if necessary. Calculator is allowed during exams. Computers and electric devices with internet services are not allowed during exams.

Academic honesty: As a student of University of Illinois at Chicago (UIC), you have agreed to abide by the University's academic honesty policy, "A Culture of Honesty," and the Student Honor Code. All academic work must meet the standards described in "A Culture of Honesty". Lack of knowledge of the academic honesty policy is not a reasonable explanation for a violation. Questions related to course assignments and the academic honesty policy should be directed to the instructor.

Disability services: Concerning disabled students, the University of Illinois at Chicago is committed to maintaining a barrier-free environment so that individuals with disabilities can fully access programs, courses, services, and activities at UIC. Students with disabilities who require accommodations for full access and participation in UIC Programs must be registered with the Disability Resource Center (DRC). Center information can be found at [http://www.uic.edu/depts/oar/campus\\_policies/disability\\_notification.html](http://www.uic.edu/depts/oar/campus_policies/disability_notification.html)

Appropriate classroom behavior: The purpose of this is to ensure the equal right of learning. No phone call is allowed during the lectures in the classroom. Cell phone should be set up to silence.

**The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.**